Day 3 & Day 4 Mastery Exam

Machine Learning Engineering — Senior Level Examination

Name: ____________________________ Date: ____________________ Score: ______ / 100

Total Score: 100 points | Duration: 90 minutes | 25 Multiple-Choice Questions (4 points each)

Select the single best answer for each question. No calculators permitted.

Section	Q1–5	Q6–10	Q11–15	Q16–20	Q21–25
Points	20	20	20	20	20
Score					

Question 1: The Revenue Forecast

You deploy a model to forecast monthly business revenue. Your evaluation metrics output an MAE of $\$5,000$ and an RMSE of $\$45,000$. What does this specifically tell the business stakeholders about the model's behavior?

A) The model is consistently off by $\$45,000$ every day. B) The model is usually off by $\$5,000$, but RMSE's sensitivity to large errors reveals it occasionally makes massive mistakes (outliers) that cost the business heavily. C) The model has a negative R^2. D) The model is perfectly homoscedastic.

Question 2: The Unbiased Estimate

A junior developer tunes a Ridge Regression model. They test `alpha = 1.0` and check the test set. They test `alpha = 10.0` and check the test set. They test `alpha = 100.0`, see the test set R^2 hit 0.92, and deploy the model. Why is this a catastrophic violation of the ML workflow?

A) Ridge requires `alpha` to be negative. B) They should have checked MAE instead of R^2. C) Validation is for decisions; the final test set is for the final unbiased estimate after decisions. By repeatedly checking the test set, they leaked data and destroyed the unbiased estimate. D) The test set should only be used for K-Fold Cross Validation.

Question 3: The Cross-Entropy Execution

You are using Logistic Regression to predict if a user will churn ($y=1$). The model calculates a probability of 0.01 (1%). In reality, the user actually churns ($y=1$). Inside the Binary Cross-Entropy (Log Loss) function, what happens?

A) The right side of the equation $(1-y)\log(1-p)$ activates, outputting zero penalty.

B) The equation outputs 0 because the threshold is 0.5.

C) The left side of the equation $y\log(p)$ activates, and because $\log(0.01)$ is a large negative number, the model suffers a massive penalty for confident ignorance.

D) The error is squared, resulting in a low penalty.

Question 4: The Fraud Threshold

You are building an anomaly detection system for credit card transactions. You calculate the probability of fraud using the Sigmoid function. Why would you manually alter the classification threshold from the standard 0.5?

A) Because 0.5 isn't always the optimal threshold. You would lower it (e.g., 0.15) to catch potential fraud earlier, minimizing business risk. B) Because Logistic Regression requires a threshold of 0.0 to minimize Log Loss. C) You increase it to 0.99 to ensure you catch more fraud. D) The Sigmoid function cannot output values below 0.5.

Question 5: The Overstuffed Model

You add 50 new, completely random noise features to a Linear Regression model. The standard R^2 goes up slightly. However, you check another metric, and it has dropped significantly. Which metric dropped and why?

A) MAE, because the errors became absolute. B) Adjusted R^2, because it penalizes unnecessary predictors in a classical linear-model setting. C) RMSE, because the errors squared. D) Log Loss, because the threshold shifted.

Question 6: The Probability Engine

In Logistic Regression, the raw linear score is calculated as $z = w^Tx + b$. What is the exact mathematical purpose of passing z through the Sigmoid function $\sigma(z) = \frac{1}{1+e^{-z}}$?

A) To calculate the Euclidean distance between w and x. B) To penalize absolute coefficients. C) So that $0 < \sigma(z) < 1$, allowing the output to be interpreted as a probability. D) To convert the probability back into Log-Odds.

Question 7: The Hidden Architecture of Classification

While the Sigmoid function outputs the probability p, what is Logistic Regression actually modeling as a linear function under the hood?

A) The Manhattan distance. B) The Mean Absolute Error. C) The log-odds, as a linear function: $\ln(\frac{p}{1-p}) = w^Tx + b$. D) The square root of the residuals.

Question 8: The Gradient Identity

Look at the formula for the weight gradient in Logistic Regression: $\frac{\partial J}{\partial w} = \frac{1}{n}X^T(p-y)$. How does this compare to the gradient formula for standard Linear Regression?

A) It requires a sigmoid derivative, making it completely different.

B) It is structurally identical, except p (probability) replaces $\hat{y}$ (linear prediction).

C) It adds a penalty term $+ 2\alpha w$.

D) It calculates distance instead of slope.

Question 9: The Danger of Capacity

You apply PolynomialFeatures(degree=25) to a server telemetry dataset. The training error drops to zero, but the validation error explodes to infinity. What is the diagnosis?

A) Increasing polynomial degree increases capacity and can produce severe overfitting. B) You forgot to use K-Fold Cross-Validation. C) The model underfit because the capacity was too low. D) You used Manhattan distance instead of Euclidean.

Question 10: The Lasso Superpower

You have a dataset with 5,000 features. You apply Lasso (L1) Regularization. When you check your model's coefficients (model.coef_), you notice that 4,800 of them are exactly 0.0. Why did this happen?

A) A bug in sklearn caused the weights to crash to zero. B) You forgot to scale the data. C) Lasso penalizes absolute coefficients, which sometimes sets coefficients to zero, acting as an automatic feature selector. D) Ridge regularization deleted the squared coefficients.

Question 11: The Regularization Tradeoff

Why would a Senior ML Engineer intentionally apply a regularization penalty to a model, knowing it will make the training error slightly worse?

A) Because Regularization trades a little training fit for better generalization. B) To convert the model into a K-Nearest Neighbors classifier. C) To increase the variance of the model. D) To speed up the Gradient Descent loop.

Question 12: The Scaling Vulnerability

You train a Ridge Regression model on raw, unscaled data containing Number of Clicks (Range: 1-10) and Server Bandwidth Bytes (Range: 1,000,000-50,000,000). The model completely ignores the bandwidth feature. Why?

A) Ridge inherently deletes large features. B) Feature scaling is particularly important when regularization penalties act directly on coefficient magnitudes. The bandwidth feature had a microscopic weight, so it was unfairly ignored by the penalty. C) Regularization strength is a hyperparameter and was set too low. D) Bandwidth requires a polynomial feature.

Question 13: The Hyperparameter Dial

You are implementing Regularization. You test $\alpha = 0.1$, $\alpha = 1.0$, and $\alpha = 10.0$. How do you know which one to pick for production?

A) Always pick 1.0 as the industry standard. B) Pick the one that drives training error to zero. C) Regularization strength is a hyperparameter and must be selected using validation. D) Pick the one that maximizes the absolute coefficients.

Question 14: The Lazy Learner

Which of the following algorithms does NOT learn a mathematical equation (like w^Tx+b) during training, but instead relies entirely on the physical geometry of the training data?

A) Logistic Regression B) Ridge Regression C) Lasso Regression D) K-Nearest Neighbors (KNN)

Question 15: The Euclidean Ruler

You are using KNN. You need to measure the straight-line distance between two points in a two-dimensional plane. Which mathematical formula is the engine using?

A) $d = \sum_{i=1}^n \vert{}x_i - y_i\vert{}$

B) $d = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$

C) $\hat{y} = \text{mode}(y_1, y_2, ... y_k)$

D) $w_i = \frac{1}{d_i}$

Question 16: The City Block

Which distance metric forces the KNN algorithm to travel along a strict grid (absolute differences) rather than a diagonal line?

A) Minkowski distance with $p=2$

B) Euclidean distance

C) Manhattan distance: $d(x,y) = \sum_{i=1}^n \vert{}x_i - y_i\vert{}$

D) Sigmoid distance

Question 17: The Master Distance Formula

The Minkowski distance is defined as $d(x,y) = \left(\sum_{i=1}^n \vert{}x_i - y_i\vert{}^p \right)^{1/p}$. What happens to this formula when you set the hyperparameter $p = 1$?

A) It becomes Euclidean distance. B) It becomes Manhattan distance. C) It triggers Ridge Regularization. D) It causes data leakage.

Question 18: KNN Voting Mechanics

You are deploying a KNN Classification model to predict if a user is a "Bot" (1) or "Human" (0). You set $k=5$. For a new user, the 5 closest neighbors are: [Bot, Human, Bot, Bot, Human]. What is the math formula the model uses to assign the final prediction?

A) $\hat{y} = \frac{1}{k}\sum_{i=1}^k y_i$

B) $\hat{y} = \text{mode}(y_1, y_2, ... y_k)$

C) $\hat{y} = \sum w_i y_i$

D) $\hat{y} = \sigma(z)$

Question 19: KNN Averaging Mechanics

You switch the system to KNN Regression to predict the continuous age of a player based on their stats. The 3 nearest neighbors are ages 20, 24, and 34. What formula does the algorithm use to predict the age?

A) The basic prediction is the average: $\hat{y} = \frac{1}{k}\sum_{i=1}^k y_i$. B) The basic prediction is the mode: $\hat{y} = \text{mode}(y_1, y_2, ... y_k)$. C) It uses Logistic Regression log-odds. D) It applies Ridge Regularization to the ages.

Question 20: The Proximity Power

In standard KNN, all k neighbors get an equal vote. How do we mathematically upgrade the engine so that a neighbor who is 0.1 miles away has more influence than a neighbor who is 5.0 miles away?

A) Use Distance-weighted KNN, where a common weighting idea is $w_i = \frac{1}{d_i}$. B) Apply Lasso Regularization to set far neighbors to zero. C) Use Minkowski $p=3$. D) Apply K-Fold Cross Validation.

Question 21: The Geometry Trap

You run KNN on a dataset with `Age=20` and `Income=100000` without applying `StandardScaler`. What is the physical result within the Euclidean distance formula?

A) The Age difference will dominate the distance. B) The income difference can completely dominate the distance, ignoring the age gap. C) The algorithm defaults to Manhattan distance. D) The model will overfit perfectly.

Question 22: The Day 4 Pipeline

According to your syllabus, what is the exact chronological implementation pattern for deploying a regularized polynomial model?

A) Generate polynomial features $\rightarrow$ Add regularization $\rightarrow$ Final test $\rightarrow$ Baseline model. B) Final test $\rightarrow$ Add regularization $\rightarrow$ Generate polynomial features. C) Baseline linear model $\rightarrow$ generate polynomial features $\rightarrow$ validate complexity $\rightarrow$ add regularization $\rightarrow$ compare validation performance $\rightarrow$ final test. D) Validate complexity $\rightarrow$ Baseline linear model $\rightarrow$ Final test.

Question 23: Cross-Validation vs Single Split

Why does K-Fold Cross-Validation provide a safer estimate of model performance than a standard `train_test_split`?

A) It automatically applies Ridge Regularization. B) Cross-validation provides multiple training/validation views of the training data, neutralizing the risk of a "lucky" or "unlucky" random data split. C) It deletes outliers using Z-scores. D) It replaces the need for a final test set.

Question 24: Diagnosing the Curve

You run the complexity curve experiment: increasing polynomial degree one step at a time and tracking training vs validation error. At Degree 2, Training Error is 10, Validation is 11. At Degree 12, Training Error is 0.5, Validation is 400. What happened?

A) The model achieved perfect capacity. B) The model underfit. C) Severe overfitting. High variance caused the engine to memorize noise. D) The learning rate was too high.

Question 25: Ridge Penalty Mechanic

What exactly does Ridge Regression add to the standard error cost function to handicap the optimization engine?

A) It penalizes squared coefficients (adding a penalty like $+ \alpha w^2$). B) It penalizes absolute coefficients. C) It multiplies the error by Log Loss. D) It adds the Minkowski distance.